AVAILABLE FOR FALL 2026 CO-OP

I build across the stack—from *circuits to systems.*

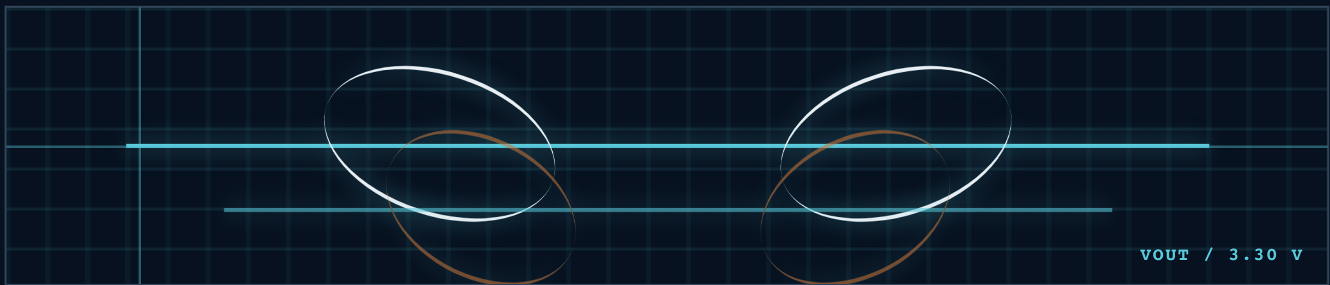
I’m **Jiacheng Wang**, an Electrical Engineering co-op student at the University of Waterloo. I design hardware, embedded systems and engineering tools that are built to be understood, tested and used.

Explore my work ↓

GitHub ↗

SYSTEM / J.WANG

● ONLINE



PCB	RTL	BOT
POWER	**VERILOG**	**ROS 2**
ALTIUM / KICAD	MAX 10 / FPGA	LIDAR / VISION

43° 28' N

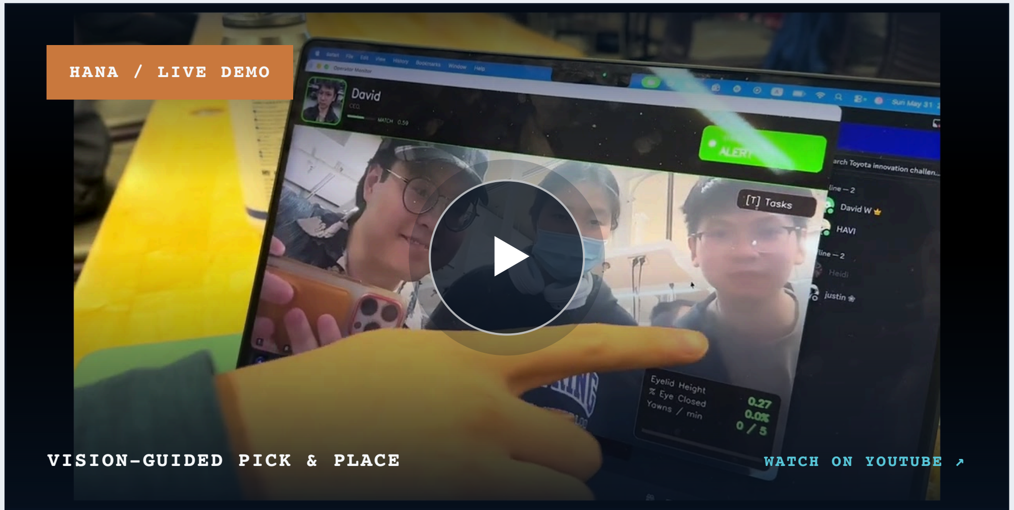
WATERLOO, ON

REV. 2026.07

01 / SELECTED WORK

Proof is in the build.

Projects spanning power electronics, RF sensing, digital design, robotics and electrical-system software.



01 COLLABORATIVE ROBOTICS · SAFETY

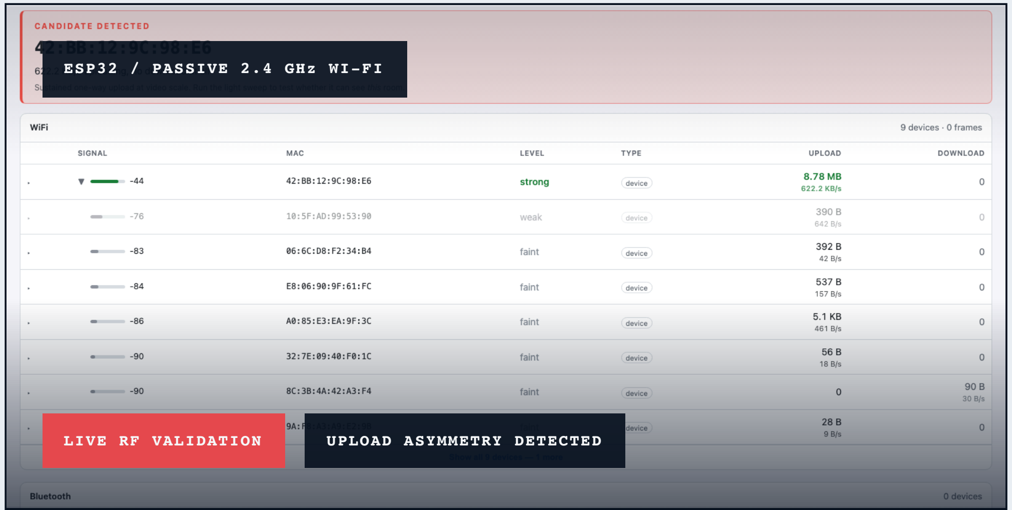
Human-Aware Robotic Arm

HANA is a vision-guided pick-and-place system that reads hand presence and operator state before delivering a part.

Best Prototype · three-zone motion scheme · automatic and manual modes

DOBOT MAGICIAN · ORBBEC · MEDIAPIPE · PYTHON

View repository ↗ · Watch demo ↗ · Presentation (PDF) ↗



02 EMBEDDED SYSTEMS · RF SENSING

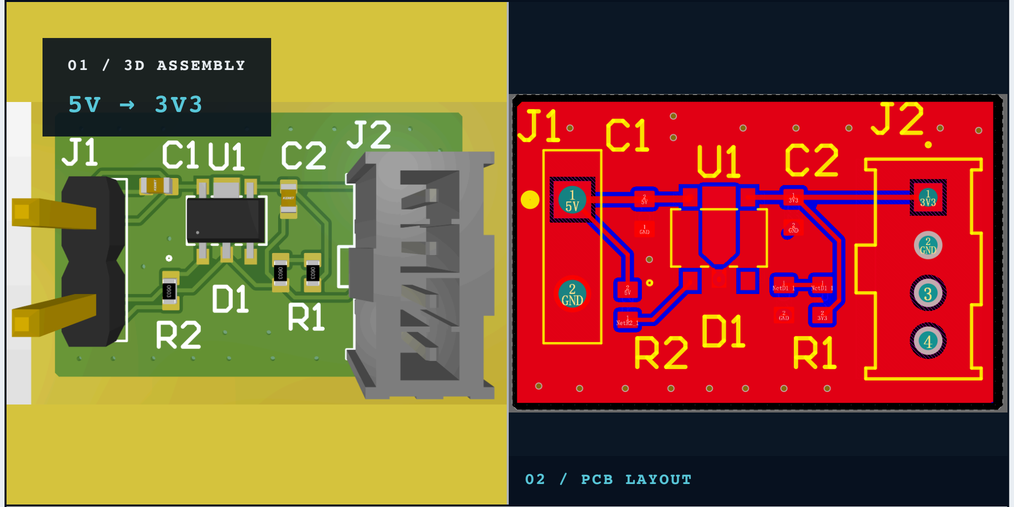
WashroomSweep

An ESP32-based privacy sweep that passively observes 2.4 GHz Wi-Fi traffic and flags upload-heavy devices before someone enters a closed space.

Passive per-device traffic analysis · 200 ms heartbeat · explicit unknown states instead of false all-clears

ESP32 · WI-FI PROMISCUOUS MODE · PYTHON · SIGNAL ANALYSIS

View repository ↗



03

PCB DESIGN · POWER

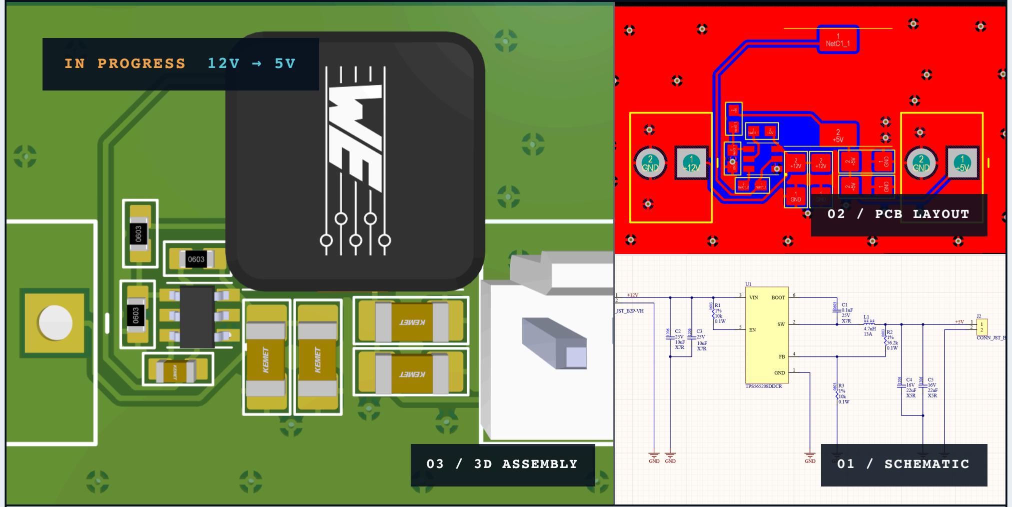
Personal LDO Regulator Board

A fabrication-ready 5 V to 3.3 V regulator board taken through the complete Altium schematic-to-layout flow.

NJM2884 · 2-layer PCB · copper pour · thermal relief · DFM review

ALTIUM DESIGNER | PCB LAYOUT | DFM

Hardware case study



04

POWER ELECTRONICS · IN PROGRESS

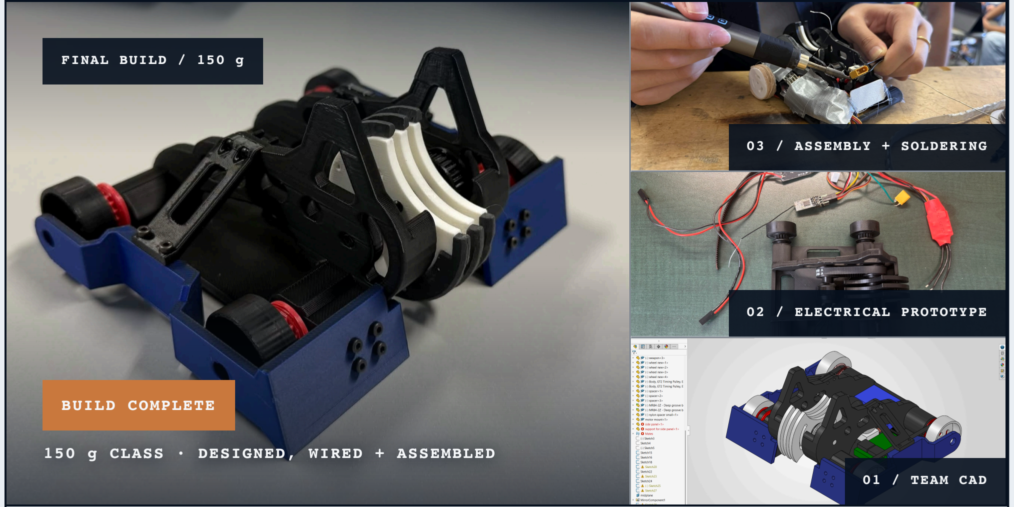
12 V to 5 V Buck Converter

An in-progress two-layer power board moving through schematic capture, component placement, routing and 3D clearance review.

Feedback network · 4.7 μH power stage · input/output filtering · two-layer PCB

ALTIUM DESIGNER | BUCK CONVERTER | PCB LAYOUT | DFM

Hardware case study



05

COMBAT ROBOTICS · COMPLETED

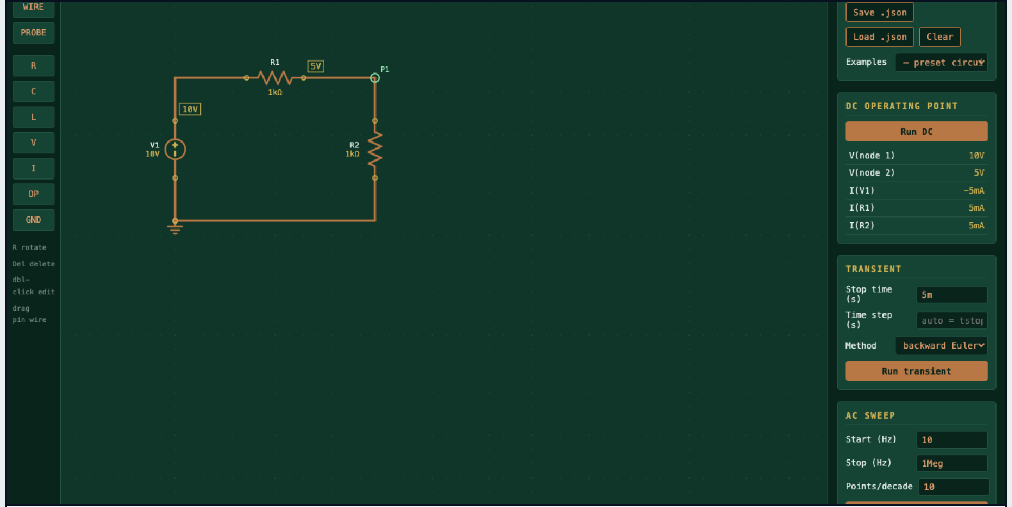
150 g Battle Robot

A completed 150 g combat robot, developed with a team from mechanical packaging through electrical integration and final assembly.

Team CAD design contributor · motor, ESC and battery selection · power-distribution planning · hands-on assembly

ELECTRICAL INTEGRATION | COMPONENT SELECTION | SOLDERING | COMBAT ROBOTICS

Hardware case study



06

ANALOG CIRCUITS · SIMULATION

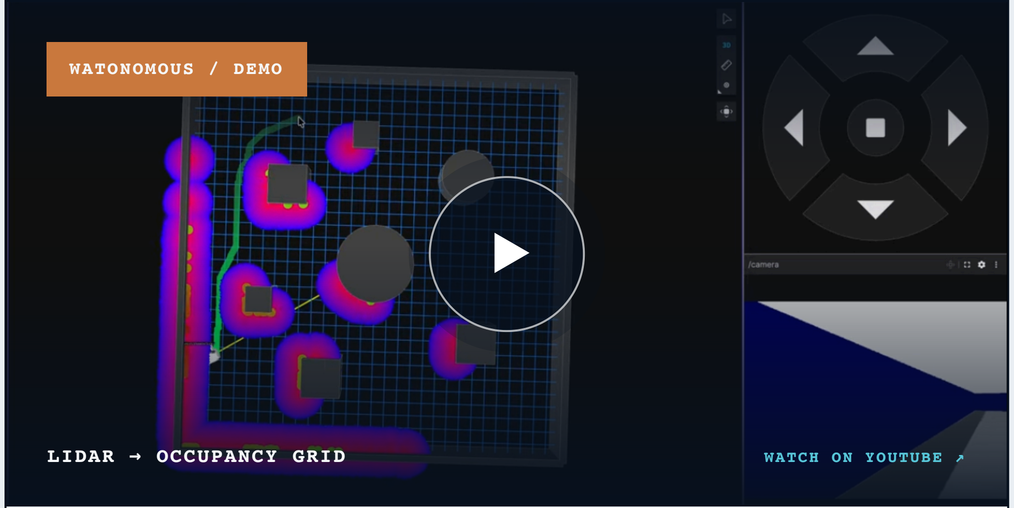
MiniSpice

A dependency-free browser circuit simulator with a hand-written Modified Nodal Analysis engine and LU solver.

DC, transient and AC analyses verified against analytical results

MNA | NUMERICAL METHODS | JAVASCRIPT

View repository ↗ Live demo ↗



07

EMBEDDED SOFTWARE · ROS 2

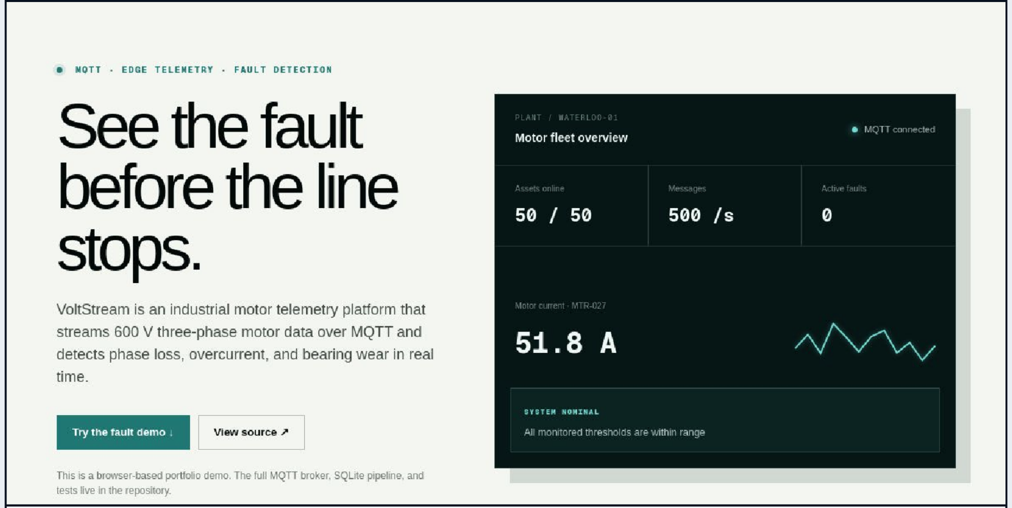
Autonomous Robot Navigation

A C++ ROS 2 node that turns raw 2D LiDAR scans into a dynamically sized occupancy-grid costmap.

2D LiDAR · dynamic occupancy grid · obstacle-cost falloff · Foxglove

C++ | ROS 2 | LIDAR | DOCKER

View repository ↗ Watch demo ↗



08

ELECTRICAL SYSTEMS · MONITORING

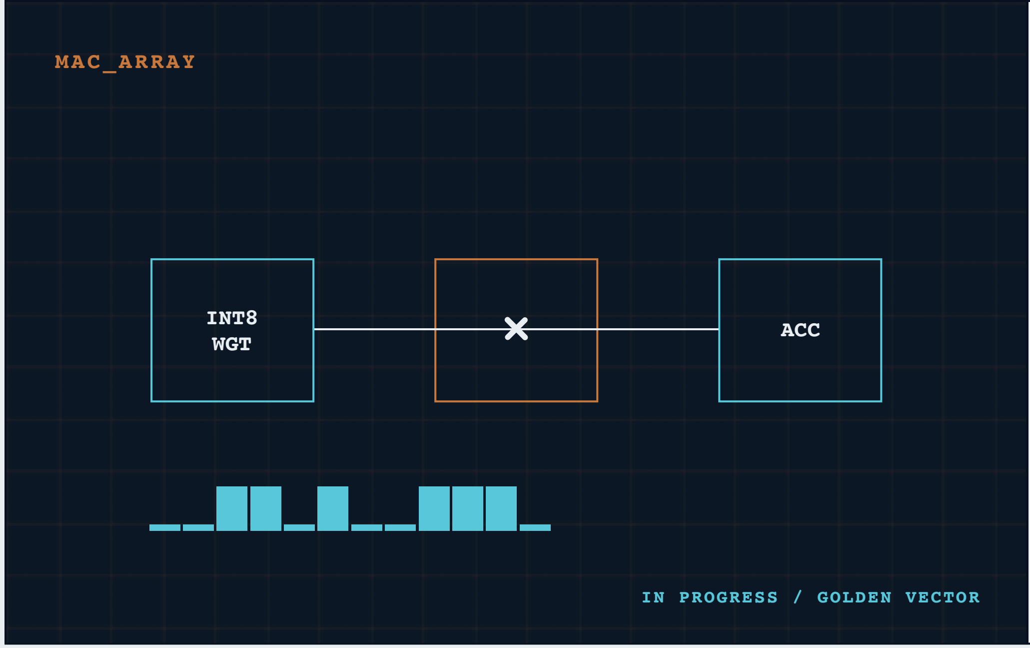
VoltStream

A real-time fault-injection lab for fleets of 600 V three-phase induction motors, with explainable fault detection.

MQTT telemetry · 50-device simulation · live diagnostics

THREE-PHASE MOTORS | MQTT | SSE

View repository ↗ Live demo ↗



09

DIGITAL DESIGN · IN PROGRESS

FPGA Neural-Network Accelerator

A from-scratch Verilog inference accelerator, built from a MAC unit toward an MNIST datapath on an Intel MAX 10.

Golden-vector co-simulation · Quartus · TinyTapeout target

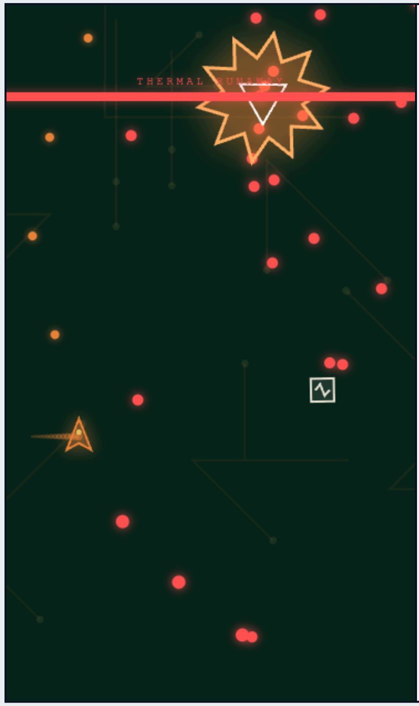
VERILOG FPGA QUARTUS PYTHON

View repository ↗

APPENDIX / OTHER BUILDS

Built for the fun of figuring it out.

Smaller software experiments that explore real-time interaction, game systems and browser-native engineering.



A / 01 BROWSER GAME · SOFTWARE EXPERIMENT

SIGNAL // Signal Defender

A PCB-aesthetic bullet-hell shooter built as one dependency-free HTML file, with six modes, bilingual controls and mobile support.

Canvas 2D · procedural Web Audio · 23 ships · seeded daily challenge

VANILLA JAVASCRIPT | CANVAS 2D | WEB AUDIO | ZERO DEPENDENCIES

View repository ↗ **Play online ↗**

A / 02 P2P MULTIPLAYER · CARD GAME

Dou Dizhu Online

A browser-based Dou Dizhu game for one to three people, with AI-filled seats and serverless peer-to-peer rooms.

WebRTC rooms · rules engine + heuristic AI · 720 simulated test rounds

JAVASCRIPT | WEBRTC | GAME AI | NODE TESTS

View repository ↗ **Play online ↗**

02 / EXPERIENCE

From schematic to flight hardware.

01

MAY 2026 — PRESENT

Electrical Team Member

Waterloo Aerial Robotics Group

Designing an in-progress 12 V buck-converter power board in Altium, from schematic capture and footprint selection through PCB layout and 3D clearance review.

02

MAY 2026 — PRESENT

Electrical Team Member

Waterloo Rocketry

Assembling and validating avionics PCBs with continuity checks, battery measurements, component-spec confirmation and PCB revision feedback.

03

JAN — APR 2026

UX/UI Designer

WE Accelerate · Magnify Access

Translated client requirements into accessible, multi-role product workflows and Figma specifications. Earned an Outstanding co-op evaluation.

Hands-on by default.

01

PROGRAMMING

- C++
- Python
- Java
- Verilog HDL

02

HARDWARE + EDA

- Altium Designer
- KiCad
- Quartus Prime
- PCB layout
- Soldering
- ESP32

03

ROBOTICS

- ROS 2
- Docker
- Foxglove Studio
- LiDAR

04

DESIGN + SIMULATION

- COMSOL Multiphysics
- Figma
- Git / GitHub

04 / ABOUT

Curious about the layer *underneath.*

Whether I'm tracing a rail on a PCB, converting LiDAR returns into a costmap, or writing a simulator from first principles, I want to understand the mechanism—not just make the demo work.

At Waterloo, I'm building that depth through Electrical Engineering coursework, design teams, and self-directed projects. I'm currently looking for a Fall 2026 co-op where I can contribute to hardware, embedded or electrical systems with a team that values careful engineering.

UNIVERSITY OF WATERLOO 2025 — 2030

BASc, Electrical Engineering (Co-op)

HAVE A HARD PROBLEM?

Let's build it.

↗